BMW Global Sales Analysis and Performance Insights (2018–2025)

Project Overview

This project analyses BMW's global sales performance from 2018–2025, focusing on vehicle sales, revenue generation, regional performance, product distribution, and electric vehicle (EV) dynamics. Using Power BI, the dataset was cleaned, modeled, and visualized to uncover key business insights and trends over time. The project integrates sales data with economic indicators such as GDP growth to provide a comprehensive understanding of factors influencing automotive performance.

Project Objectives

1. **Analyse Sales Trends:**
   To study monthly and yearly vehicle sales patterns and identify seasonal variations.
2. **Evaluate Regional Performance:**
   To analyse revenue contribution across regions such as China, Europe, USA, and Rest of World.
3. **Assess Product Performance:**
   To identify top-performing BMW models and evaluate sales distribution.
4. **Analyse EV Market Dynamics:**
   To examine EV model performance and contribution to total vehicle sales.
5. **Understand Economic Impact:**
   To evaluate how GDP growth influences sales trends and revenue patterns.

Data Sources

- **Source:** BMW Global Sales Dataset (Kaggle)
- **Time Period:** 2018–2025

- **Domain:** Automotive / Sales Analytics

The dataset includes sales data, revenue, model details, regional distribution, EV sales, and external indicators such as GDP growth and fuel price index

---

Problem Statement

- To identify which regions generate the highest revenue
- To analyse which BMW models contribute most to total sales
- To determine whether sales follow seasonal patterns
- To evaluate the competitiveness of the EV segment
- To assess the impact of economic conditions on vehicle sales
- To analyse the contribution of EV sales to total market share

---

Attribute Details

Attribute Name	Data Type	Description
Year	Integer	Sales year (2018–2025)
Month	Integer	Month number
Month Name	Text	Month name
Region	Text	Sales region
Model	Text	BMW vehicle model
Units_Sold	Whole Number	Total vehicles sold
Revenue_EUR	Currency	Revenue generated
EV_Sales	Whole Number	Electric vehicle sales
GDP_Growth	Decimal	Economic indicator

Fuel_Price_Index	Decimal	Fuel trend indicator
Avg_Price_EUR	Currency	Average vehicle price
BEV_Share	Decimal	EV share indicator

Tools and Technologies

Microsoft Excel

Used for initial data collection and preprocessing. Power Query was used to clean, transform, and structure the dataset.

Microsoft Power BI

Used for data modeling, DAX calculations, and dashboard development. Interactive visualizations were created to analyse trends and performance.

Data Preprocessing (Power Query)

Data Cleaning

- Removed duplicate records
- Handled missing values
- Standardized column formats

Data Transformation

- Created calculated columns (average price, EV metrics)
- Classified sales into categories

Data Integration

- Combined all cleaned data into a single dataset for analysis

Data Modelling and DAX (Power BI)

Data Modelling

A structured data model was created using a star schema approach, where the cleaned dataset acts as the fact table and is connected with dimensions such as Year, Month, Region, and Model. This design enables efficient filtering, aggregation, and multi-dimensional analysis across different business perspectives. The inclusion of sales metrics, EV indicators, and economic variables supports dynamic insights and enhances the analytical capability of the dashboard.

Dataset Fields (Power BI Model)

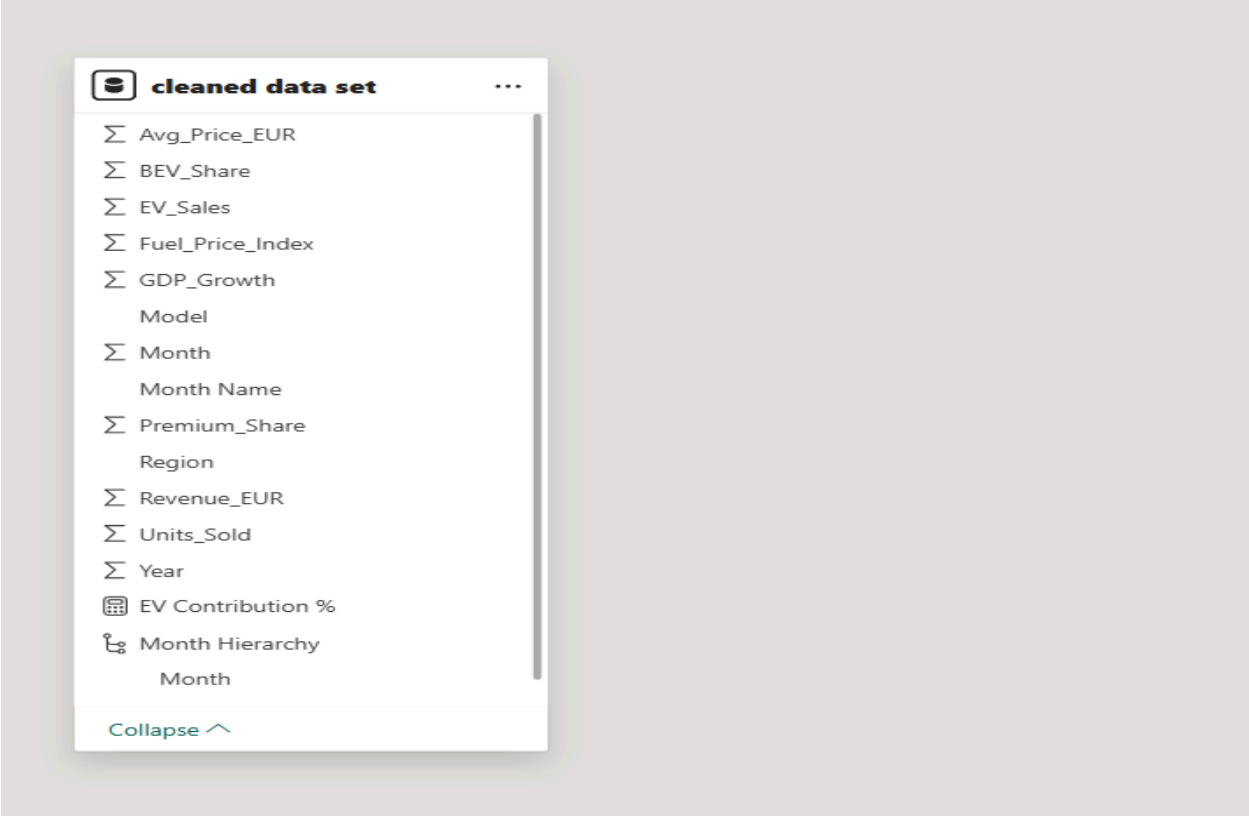

Fig 2 :Power BI Data Model – Cleaned Dataset Fields and Measures

Dataset Explanation

The dataset consists of key fields used for analysis:

- **Units_Sold** and **Revenue_EUR** for sales and revenue analysis
- **EV_Sales** and **BEV_Share** for electric vehicle insights
- **GDP_Growth** and **Fuel_Price_Index** for economic impact analysis
- **Avg_Price_EUR** for pricing analysis
- **Region, Model, Year, Month** for filtering and segmentation

Additionally, calculated measures such as **EV Contribution %** were created using DAX to evaluate the proportion of electric vehicle sales relative to total vehicle sales.

DAX Measures

Measure Name	Formula	Purpose
Total Vehicles Sold	`SUM('cleaned data set'[Units_Sold])`	Calculates total vehicles sold
Total Revenue	`SUM('cleaned data set'[Revenue_EUR])`	Calculates total revenue
Average Vehicle Price	`DIVIDE([Total Revenue], [Total Vehicles Sold])`	Computes average vehicle price
EV Contribution %	`DIVIDE(SUM('cleaned data set'[EV_Sales]), SUM('cleaned data set'[Units_Sold]))`	Measures EV share in total sales

Analysis and Visualizations (Power BI)

To address the identified problem statements, a series of interactive Power BI visualizations were developed. Various charts such as line charts, bar charts, donut charts, and KPI cards were used to analyse BMW's global sales performance, highlight trends, and provide actionable insights.

---

1. Sales Trend Analysis

To analyse monthly and yearly sales performance, a **line chart** was used to visualize vehicle sales across different years.

- The chart highlights **seasonal variations** in demand
- Identifies **peak and low sales periods**
- Helps in understanding **cyclical sales patterns over ti**

---

2. Model Performance Analysis

To evaluate product performance, a **bar chart** was used to compare total units sold across BMW models.

- Identifies **top-performing models**
- Shows **balanced distribution across product portfolio**
- Helps assess **product-level contribution to total sales**

---

3. Regional Revenue Analysis

To analyse geographic performance, a **column chart** was used to compare revenue across regions.

- Highlights **China as the highest revenue contributor**
- Enables comparison between **Europe, USA, and Rest of World**
  Supports **regional market performance evaluation**

---

4. EV Market Share Analysis

A **donut chart** was used to represent the distribution of electric vehicle models.

- Shows contribution of **iX, i4, and MINI within EV segment**
- Highlights **competitive distribution among EV models**
- Helps understand **EV product positioning**

---

5. Economic Impact Analysis

A **dual-axis line chart** was used to compare revenue trends with GDP growth.

- Demonstrates relationship between **economic conditions and sales performance**
- Provides insight into **external factors influencing demand**
- Enhances analytical depth beyond descriptive analysis

---

6. KPI Metrics

KPI cards were created to display key performance indicators:

- **Total Vehicles Sold: 24.52M**
  This indicates that BMW sold approximately 24.52 million vehicles globally during the analysis period, reflecting strong market demand and consistent sales performance across regions.
- **Total Revenue: €1.57T**
  The total revenue generated is €1.57 trillion, showing BMW's strong financial performance and its ability to generate high-value sales in the global automotive market.
- **Average Vehicle Price: €45.04K**
  The average price per vehicle is €45.04K, indicating BMW's positioning in the premium automotive segment and maintaining stable pricing across its product portfolio.

- **EV Contribution: 11.1%**

Electric vehicles contribute 11.1% of total sales, indicating that EV adoption is still in the early growth stage. This suggests strong future potential for expansion in the EV segment as demand increases.

BMW Sales Dashboard

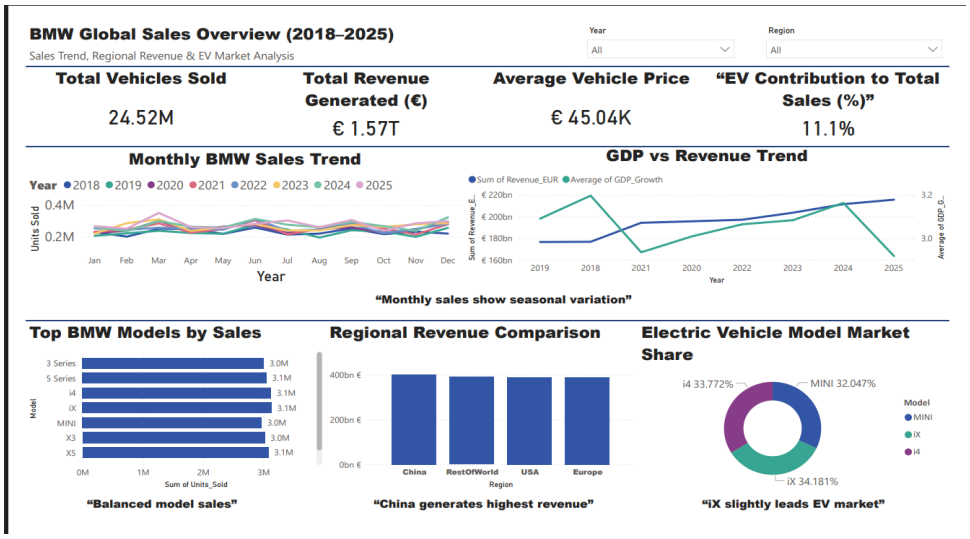

Figure 3: BMW Global Sales Dashboard Overview

Description:

The dashboard provides a comprehensive view of BMW's global sales performance, integrating key metrics, trends, and comparisons into a single interactive interface. It enables users to analyse sales patterns, regional performance, and EV market dynamics effectively.

Insights

- China is the highest revenue-generating region, indicating strong demand and market dominance in the automotive sector

- Sales exhibit seasonal variation, reflecting cyclical demand patterns across months
- BMW maintains a balanced product portfolio with consistent performance across models
- EV models (iX, i4, MINI) show competitive distribution within the EV segment
- EV contribution (~11%) indicates early-stage adoption with potential for future growth
- GDP growth trends influence revenue performance, highlighting the impact of economic conditions on sales

---

Conclusion

The integration of Excel and Power BI enabled a comprehensive analysis of BMW's global sales performance from 2018 to 2025. The dataset was cleaned, transformed, and modeled to create an interactive dashboard that provides meaningful insights into sales trends, regional performance, and product distribution.

The analysis highlights strong market performance driven by China, seasonal demand patterns, and a balanced product strategy. While EV adoption is growing, it remains a smaller portion of total sales, indicating future growth opportunities. Additionally, economic indicators such as GDP play a significant role in influencing sales trends.

Overall, the project demonstrates how data analytics can support data-driven decision-making and strategic business planning in the automotive industry.